# 第七章 死锁

1. **Consider the traffic deadlock depicted in Figure 1.**
   **a. Show that the four necessary conditions for deadlock indeed hold in this example.**
   **b. State a simple rule for avoiding deadlocks in this system.**

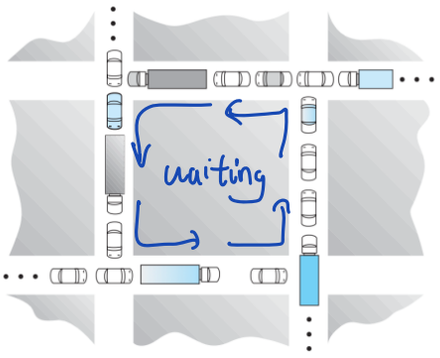



**Figure 1. Traffic deadlock**

a. ① Mutual exclusion: The roads are resources that can't be shared. Only one thread at a time can use a resource.

② Hold and wait: The car holds the road it currently occupies while waiting to acquire another road to move into. Thread holding at least one resource is waiting to acquire additional resources held by other threads.

③ No preemption: It's impossible to remove a car from a road forcibly, it can only leave by itself. Resources are released only voluntarily by the thread holding the resource, after thread is finishing with it.

④ Circular wait: There is a waiting cycle like picture shows.

b. Don't enter the road unless your entire path through it is clear. It means to prevent system from reaching an unsafe state which may result in a deadlock.

2. **Consider the following snapshot of a system:**

| | ***Allocation*** A B C D | ***Max*** A B C D | ***Available*** A B C D |
|---|---|---|---|
| $P_0$ | 0012 | 0012 | 1520 |
| $P_1$ | 1000 | 1750 | |
| $P_2$ | 1354 | 2356 | |
| $P_3$ | 0632 | 0652 | |
| $P_4$ | 0014 | 0656 | |

**Answer the following questions using the banker's algorithm:**

**a. What is the content of the matrix Need?**

**b. Is the system in a safe state?**

**c. If a request from process $P_1$ arrives for (0,4,2,0), can the request be granted immediately?**

a. Need = Max - Allocation

| | A | B | C | D |
|---|---|---|---|---|
| $P_0$ | 0 | 0 | 0 | 0 |
| $P_1$ | 0 | 7 | 5 | 0 |
| $P_2$ | 1 | 0 | 0 | 2 |
| $P_3$ | 0 | 0 | 2 | 0 |
| $P_4$ | 0 | 6 | 4 | 2 |

b. $P_0 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_1$

(1,5,3,2) (2,8,8,6) (2,14,11,8) (2,14,12,12) (3,14,12,12)

It's safe!

c. Request ≤ Need, ≤ Available

After allocating the request, Available = (1, 1, 0, 0)

$P_0 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_1$

(1,1,1,2) (2,4,6,6) (2,10,9,8) (2,10,10,12) (3,10,10,12)

It's safe, so the request can be granted immediately.